ELEC 271– DIGITAL SYSTEMS

Course Syllabus – Fall 2026

This is your course syllabus. Please download the file and keep it for future reference.

TEACHING TEAM

LAND ACKNOWLEDGEMENT

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory. See: http://www.queensu.ca/encyclopedia/t/traditional-territories

INCLUSIVITY STATEMENT

Queen's students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: https://www.queensu.ca/vpcei/support/overview.

COURSE INSTRUCTOR

Muhammad Saad Asad, PhD
Dept. of Electrical and Computer Engineering

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Office hours: By appointment

TEACHING ASSISTANTS

Teaching Assistants	Last name, first	Email
Graduate	Ghojavand, Hossein	23rj24@queensu.ca
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Undergraduate	Bingham, Mekai	
	Edwards, Taylor	
	Kourkopolous, Luka	
	Letchumanan, Nagappan	
	Odorico, Nicolas	
	Starosta, Hubert	
	Wen, William	

COURSE SUCCESS

Course success requires that you pay close attention to announcements on OnQ. Please subscribe for receiving notification of announcements in case of changes to important course information. Midterms locations and times are subject to change, please check back frequently to ensure correct location and time for evaluations in this course.

CLASSES WILL BE DELIVERED IN PERSON

Tentative Midterm information, please check OnQ the week before and of the midterm for any changes or announcements.

Midterm:

-**October 29th**, during class period.
-Location TBD

Lectures:

Check your solus schedule for lecture times and locations

Tutorials:

Check your solus schedule for tutorial times and locations

Labs:

Check your lab section! Only attend the lab you're scheduled for or there will not be equipment you need for the lab available to you.

Follow your SOLUS schedule if it does not agree with the information here.

005, 006, 009, and 010 start in week 2 (week of Sept 14th)
007, 008, 011, and 012 start in week 3 (week of Sept 2st)

Note that there are only 4 labs per student. You will not have a lab every week, and some periods you don't have your regular bi-weekly lab. The labs are suspended near the midterm to allow time to study.

Mondays 6:30 - 9:30pm BMH 212, 214
Tuesdays 6:30 - 9:30pm BMH 212, 214

Due to hardware availability, you are only allowed to attend your predetermined lab session. Any accommodations must be approved and arranged with the teaching team.

ELEC 271 (F 3-0.5-0.5 4.00)

Course Description

Boolean algebra applied to digital systems; logic gates; combinational logic design; electronic circuits for logic gates; arithmetic circuits; latches and flipflops, registers and counters; synchronous sequential logic and state machine design; implementation in programmable logic chips. (0/0/0/21/27).

Mode of Instruction

The delivery mode of this course is in-person.

Pre-Requisite Knowledge

Prerequisites: APSC 171, APSC 172, APSC 174 or permission of the instructor

Course Learning Outcomes (CLO)

By the end of this course, students should be able to:

CLO	Description	Indicator[1]
CLO 1	Represent combinational logic building blocks such as multiplexers, encoders, and decoders in algebraic form, in schematic form, and in the syntax of a hardware design language for computer-aided logic synthesis.	KB-ES (Digital Systems and Architecture)
CLO 2	Represent latches and flip-flops for storing information in schematic form and in hardware-design-language syntax and describe the timing behavior of flip-flops.	KB-ES (Digital Systems and Architecture)

[1] For further information on graduate attributes and indicators, please click here.

CLO	Description	Indicator[1]
CLO 3	Optimize a combinational logic function with Karnaugh maps, both with and without don't-care valuations.	KB-ES (Digital Systems and Architecture)
CLO 4	Design a finite-state machine from a given state diagram by directly deriving and optimizing the next-state and output logic, and by generating a behavioral description in a hardware design language for computer- aided logic synthesis.	DE-Solutions

COURSE EVALUATION

Assessment Weighting

Assessment Tool	Due Date (before 23:59 ET)	Weight	Alignment with CLOs
Midterm	Week 7, **Thursday October 29th**	**30%**	
Labs	In person in the lab	**30%**	
Final Exam	Please see the Final Exam schedule posted by OUR Exams Office	**40%**	**All**
		100%	

ASSESSMENT DESCRIPTIONS

Midterm

There is a single midterm in this course on **Thursday October 29th** during class. This is located between your labs to help you have time to study. All sections of the course write the midterm on the same day. Locations of the midterm are provided above with the course schedule. Please check the course website to confirm the location and time of the midterm.

Problem Sets

Problem sets will be provided to practice the material presented in the course. They are **not marked and do not account for any part of your grade**. More details about these assignments can be found in onQ.

Final Exam

The final exam is closed book. Students must write their exam on the day and time scheduled by the University. You should not schedule vacations, travel, etc. during the exam period. The Term and Session Dates will indicate the final exam period session dates in each term.

GRADING

All assessments in this course will receive numerical percentage marks. The final grade you receive for the course will be derived by converting your numerical course average to a letter grade according to the established Grade Point Index.

Any missed assessments (midterm, labs, etc.) that are missed with a valid absence (see https://engineering.queensu.ca/Current-Students/absences-accommodations/academic consideration.html) **will have grades distributed to other written assessments**. For a missed lab, the weighting will shift to other laboratories, for midterms, the weighting will be shifted to the final exam. Please note the list of accepted absences listed on the engineering website. All absences need to be approved at the faculty level and recorded by the course coordinator.

Feedback on Assessments

The teaching team will provide feedback on graded activities. You can expect feedback on your assessments within two weeks of the due date or one week before the feedback is needed for other assessments (tests/quizzes), whichever is sooner.

Accessing Your Final Grade

final grades will show on SOLUS. Official transcripts showing final grades will be available on the Official Grade Release Date. Please note that in official transcripts, a mark of IN (incomplete) is considered a grade, and your transcript is released with this grade.

COURSE MATERIALS

Required Textbook

ELEC 271 Course textbook available from the campus bookstore: Fundamentals of Digital Logic with Verilog Design ISBN: 9780073380544

Cost: $100

Other Material

All other course material is accessible via OnQ.

Required Calculator

A Casio 991 is required. **ONLY** this type of non-programmable, non-communicating calculator will be allowed during tests and exams.

Suggested Time Commitment

This course represents a study period of one term spanning 12 weeks. Learners can expect to invest on average 9-11 hours per week in this course. Learners who adhere to a predetermined study schedule are more likely to successfully complete the course

COURSE COMMUNICATION

QUESTIONS ABOUT COURSE MATERIAL

Communication policy:

Please note that this policy is essential to follow to get prompt replies from the teaching team. Due to the large enrollment of this course, following this policy will ensure fairness to students in communications with the instructor. If you need to ask a question of one of the Teaching Assistants in the course, please email them directly.

If you have feedback on the course, please talk to your student course representative who will collect anonymize and relay that feedback to the instructor.

Steps to communication with the professor:

1. Check if the information you are asking for is already in the syllabus or posted on onQ.
2. Determine if the answer to the question you may have would be common and useful to other students. If so, please post it on the communications page in OnQ where the TAs and instructor will be monitoring the Q&A board. If the question is personal or sensitive in nature, do not post it on OnQ
3. Email your section coordinator (TA) with the following at the start of the subject line [ELEC271]: . They will categorize and triage all email requests. Only requests with the correct subject line beginning will be easily caught, therefore if you do not include the required subject line, you may not get a response. a. If the topic of your email is sensitive please let the section coordinator know and they will email the professor to let them know to expect an email directly from you. Email on sensitive topics can then be sent to the professor directly.
4. You will then get an email response from the appropriate teaching team member (it may be a TA or directly from the instructor).

Questions or comments regarding the course material that can be of benefit to other students should be posted in the Q&A forum on the class website. The instructor, TAs, and students are encouraged to answer these questions directly in the discussion forum for the benefit of everyone in the course.

COURSE ANNOUNCEMENTS

The instructor will routinely post course news in the Announcements section on the main course homepage on OnQ.

Office Hours

In addition to interaction in the Q&A discussion forums, you will have the opportunity to interact with either a TA or the instructor during office hours. The instructor will provide a schedule of availability at the beginning of the term.

Confidential Matters

If you have a confidential matter you would like to discuss with your instructor, their contact details are on the first page of this document. Expect email replies within 72 hours, this is to give time to investigate a possible solution to the matter. If you would like a response indicating the receipt of your email, please indicate that in your correspondence and acknowledgement will be sent, with a goal of being within 1 business day.

Absences (Academic Considerations) and Missed Assignments

Students who miss assignments, quizzes, classes, labs, tutorials or other course deliverables due to unforeseen personal circumstances outside of their control can apply for Academic Consideration. For information on academic considerations due to extenuating circumstances, please review the information on the Smith Engineering website. Note that unacceptable reasons include extracurricular activities, work, travel plans, scheduling conflicts, generally being behind on schoolwork, etc. Do not schedule travel during midterms and final exams, as travel and work are not acceptable reasons for granting academic considerations.

Any missed assessments (midterm, labs, etc.) that are missed with a valid absence (see https://engineering.queensu.ca/Current-Students/absences-accommodations/academic consideration.html) **will have grades distributed to other written assessments**. For a missed lab, the weighting will shift to other laboratories, for midterms, the weighting will be shifted to the final exam. Please note the list of accepted absences listed on the engineering website. **All absences need to be approved at the faculty level**.

Late Policy

Laboratories in this course are due at the end of your laboratory session in which they were assigned. As such, **no late work on laboratories is allowed as your work must be checked in the lab period.**

Requesting Academic Consideration Related to Group Work

Group submissions may pose challenges when one or more members of the group experience extenuating circumstances.

In the unlikely event that further assignment work is provided, any applicable late penalties will be described in the details for each assessment. In the event of extenuating circumstances, you must follow the policies for requesting an academic consideration (please see below). Note that unacceptable reasons include extra-curricular activities, travel plans, being generally behind on schoolwork, etc. In the absence of an approved

consideration request, the normal late penalty will apply as described in the assignment or any course/departmental policies.

STANDARD QUEEN'S AND SMITH ENGINEERING POLICIES

NETIQUETTE

In this course, you may be expected to communicate with your peers and the teaching team through electronic communication. You are expected to use the utmost respect in your dealings with your colleagues or when participating in activities, discussions, and online communication.

The following is a list of netiquette guidelines. Please read them carefully and use them to guide your online communication in this course and beyond.

1. Make a personal commitment to learn about, understand, and support your peers.
2. Assume the best of others and expect the best of them.
3. Acknowledge the impact of oppression on the lives of other people and make sure your writing is respectful and inclusive.
4. Recognize and value the experiences, abilities, and knowledge each person brings.
5. Pay close attention to what your peers write before you respond. Think through and re-read your writings before you post or send them to others.
6. It's alright to disagree with ideas, but do not make personal attacks.
7. Be open to be challenged or confronted on your ideas and challenge others with the intent of facilitating growth. Do not demean or embarrass others.
8. Encourage others to develop and share their ideas.

STUDENT CODE OF CONDUCT

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's. It outlines the activities and behaviours that could be considered Non-Academic Misconduct (NAM). The Code also describes the NAM process and the sanctions that could be imposed on a student found responsible for a violation. All students should be familiar with the Student code of conduct and related policies on sexual violence prevention and response and harassment and discrimination prevention and response.
https://www.queensu.ca/nonacademicmisconduct/policies

COPYRIGHT

Course materials created by the course instructor, including all slides, presentations, synchronous and asynchronous course recordings, handouts, tests, exams, and other similar course materials, are the intellectual property of the instructor. It is a departure from

academic integrity to distribute, publicly post, sell or otherwise disseminate an instructor's course materials or to provide an instructor's course materials to anyone else for distribution, posting, sale or other means of dissemination, without the instructor's ***express consent***. A student who engages in such conduct may be subject to penalty for a departure from academic integrity and may also face adverse legal consequences for infringement of intellectual property rights and, with respect to recordings, potentially privacy violations of other students.

ACADEMIC INTEGRITY

As an engineering student, you have made a decision to join us in the profession of engineering, a long-respected profession with high standards of behaviour. As future engineers, we expect you to behave with integrity at all times. Please note that Engineers have a duty to:

- Act at all times with devotion to the high ideals of personal honour and professional integrity.
- Give proper credit for engineering work

The standard of behaviour expected of professional engineers is explained in the Professional Engineers Ontario Code of Ethics. Information on policies concerning academic integrity is available in the Queen's University Code of Conduct, in the Senate Academic Integrity Policy Statement, on the Smith Engineering website, and from your instructor.

Departures from academic integrity include plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, and collaboration, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the University.

In the case of online or remotely proctored exams, impersonating another student, copying from another student, making information available to another student about the exam questions or possible answers, posting materials to online services, communicating with another person during an exam or about an exam during the exam window, or accessing unauthorized materials, including internet sources and using unauthorized materials, including smart devices, are actions in contravention of academic integrity.

PLAGIARISM/USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (GENAI) TOOLS:

Students must submit their own work and cite the work that is not theirs. Generative AI writing tools such as ChatGPT may be permissible if explicitly noted in the assessment or lab instructions. In these cases, be sure to cite the material that they generate and

communicate that to a TA overseening the lab. Any other use constitutes a Departure from Academic Integrity.

Invalid Exams

An exam may be declared invalid in case of an interruption in an in-person examination; if the instructions in a remote or online exam were not followed; if the student uploads wrong materials; or if a situation arises where the integrity of the exam cannot be verified. If an exam is declared invalid, the student may be granted a re-write.

ACADEMIC AND STUDENT SUPPORT

Queen's has a robust set of supports available to you, including the Library, Student Academic Success Services (Learning Strategies and Writing Centre), and Career Services. Learners are encouraged to visit the Smith Engineering Current Students web portal for information about various other policies such as academic advisors, registration, student exchanges, awards and scholarships, etc. Students are also encouraged to review the information that is available in the EngQ Hub, posted in onQ.

Absences - Academic Considerations

Academic Consideration is a process by which students request validated time away from their academics when they are experiencing an extenuating circumstance or personal circumstances beyond a student's control that interfere with their ability to complete academic requirements.

For additional information and to apply for academic consideration, please review the information on the Smith Engineering's Academic Consideration (Absences) website.

Absences – Requests for Excused Absences for Significant Events (REASE)

Students participating in a Varsity Athletics event, the student reserve forces, a non-varsity event at the provincial, national, or international level, or an event to which you were invited as a distinguished guest can submit a Request for Excused Absence for Significant Event following the REASE process outlined on the Student Affairs webpage.

Students participating in Design Competitions supported by Smith Engineering can submit a Request for Excused Absence for Significant Events (REASE) following the process outlined on the Smith Engineering's Academic Consideration (Absences) website.

All Requests for Excused Absences for Significant Events must be submitted to the Faculty a minimum of 2 weeks prior to the event start date. For additional information and to submit a REASE application, please review the information on the Smith Engineering's Academic Consideration (Absences) website.

Academic Accommodations

Academic Accommodations are available when students need an adjustment to their learning environment due to a reason related to a Human Rights protected ground. Below there is information on the most common Academic Accommodations requested by students. For more information, please see Smith Engineering's Academic Accommodation website.

Religious Accommodations

Students in need of accommodation for religious observance are asked to complete the Religious Observance and Academics Activities form available on Smith Engineering's Academic Accommodation webpage within a week of receiving their course syllabus (each term).

- Alternative assignments are considered a "reasonable accommodation" under the Ontario Human Rights Code
- Please refer to the Queen's Multi-faith Calendar for all of this year's faith dates. Requests for Religious Accommodation will be verified based on the dates in this calendar. If you are requesting dates not listed in this calendar, you may be asked for additional documentation
- Students with questions about their rights and responsibilities regarding religious accommodations can contact chaplain@queensu.ca or Smith Engineering Academic Accommodation and Consideration Advisor at engineering.aac@queensu.ca

Further information can be found on the Smith Engineering's Academic Accommodation webpage

Other human-rights based Accommodations

Engineering students who need accommodation on Human Rights grounds are asked to contact, in confidence, the Smith Engineering Academic Accommodation and Consideration Advisor at mailto:engineering.aac@queensu.ca or to call 613-533-6000 x 78139.

Accommodations for Disabilities

Queen's University is committed to working with students with disabilities to remove barriers to their academic goals. Queen's Student Accessibility Services (QSAS), students with disabilities, instructors, and faculty staff work together to provide and implement academic accommodations designed to allow students with disabilities equitable access to all course material (including in-class as well as exams). If you are a student currently experiencing barriers to your academics due to disability related reasons, and you would like to understand whether academic accommodations could support the removal of those barriers, please visit the QSAS website to learn more about academic accommodation.

To start the registration process with QSAS, click the "**Access Ventus"** button found on the Ventus student portal.

Ventus is an online portal that connects students, instructors, Queen's Student Accessibility Services, the Exam's Office, and other support services in the process to request, assess, and implement academic accommodations. To learn more about Ventus, visit A Visual Guide to Ventus for Students:
https://www.queensu.ca/ventus-support/students/visual-guide-ventus-students

For questions or assistance with requesting Academic Consideration or Accommodation, contact the Smith Engineering Academic Accommodation and Consideration Advisor at engineering.aac@queensu.ca

Every effort has been made to provide course materials that are accessible. For further information on accessibility compliance of the educational technologies used in this course, please consult the links below.

Student Wellbeing

Success in Engineering includes taking care of your wellbeing. If you're experiencing stress, anxiety, burnout, personal challenges, or you're simply not sure where to turn, EngWell is here to help.

EngWell provides confidential support through our Wellness Advisor, Embedded Counsellors, mental health programming, and referrals to academic and campus resources. Whether you need short term guidance, help navigating university supports, or someone to talk to, we encourage you to reach out early.

Educational Technology

Educational Technology	Accessibility Compliance Information
onQ (Brightspace Learning Management System by D2L)	https://www.d2l.com/accessibility/standards/
MS-Teams	https://support.microsoft.com/en-us/office/accessibility-support-for-microsoft-teams-d12ee53f-d15f-445e-be8d-f0ba2c5ee68f
Zoom	https://zoom.us/accessibility

If you find any element of this course difficult to access, please discuss with your instructor how you can obtain an accommodation.

Technical Support

Some basic comfort levels with basic hardware and software skills are required for this course. If you require technical assistance, please contact Technical Support.